DE–759 **13**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2014.

DIGITAL COMPUTER FUNDAMENTALS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Convert the following

   (a) $11001.101_{10} \Rightarrow (?)_8 \Rightarrow (?)_{16}$

   (b) $5073.625_8 \Rightarrow (?)_2 \Rightarrow (?)_{16}$.

2. Discuss on Sum of Products and Products of Sums with examples.

3. Explain DeMorgan's Law with suitable example.

4. What is half adder? Design an half adder and implement it using universal gates.

5. (a) Define binary logic.

   (b) How negative numbers are stored in a digital computer?

6. What is a sequential circuit? Discuss with an example.

7. Write short notes on accumulator register.

8. Discuss about the working of computer memory.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe number systems in detail, with examples.

10. (a) With an example, discuss the 4 variable Karnaugh map simplification.

    (b) Convert the following expression into product–of–sum form $A+A'B+A'C'$.

11. Discuss about the operation of multiplexer and demultiplexer.

12. Explain addition and subtraction in a parallel arithmetic element.

13. What are flip flops? Explain Master-Slave flip flop in detail.

14. With necessary diagrams, discuss the functions of binary counters and BCD counters.

15. Illustrate on the study of generation of computers and classification.

DE–760 **14**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2014.

C AND DATA STRUCTURES

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Explain the various data types available in C.

2. Explain the Input and Output functions with example.

3. Define array. Explain the one-dimensional and two-dimensional arrays.

4. Explain any four string functions with example.

5. Define pointer. Discuss about the advantages of pointers.

6. Write a C program to generate the pay slip of employees using files.

7. Define stack. Explain the structure of a stack.

8. What is binary tree? Explain in detail about the operations on binary trees.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the various operators available in C with suitable example.

10. (a) Explain the switch statement with example.

(b) Write a C program to check whether a given number is prime number or not.

11. (a) Explain call by value and call by reference with example.

(b) Explain about macros with suitable examples.

12. (a) What is a file? Explain the various file opening modes.

(b) Discuss about array of pointers with suitable example.

13. (a) Define structure. Explain about defining structures and accessing structure members.

(b) Write briefly about dynamic memory allocation.

14. (a) Explain the queue operations.

(b) Write a brief note on abstract data types.

15. (a) Explain about singly linked list.

(b) Explain about doubly linked list.

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DE–761 **15**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2014.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Using the Statements :

   R : Ram is Rich

   H : Ram is Happy

   Write the following statements in symbolic form

   (a) Ram is poor but happy

   (b) Ram is rich or unhappy

   (c) Ram is neither rich nor happy

   (d) Ram is poor or he is both rich and unhappy.

2. Define Tautology and contradiction with examples.

3. Show that $(A + B) + C = A + (B + C)$, for any three sets A, B, C, where $A + B = (A - B) \cup (B - A)$.

4. (a) Define 1-1 and onto function.

   (b) Show that the functions f and g which both from $\mathbf{N} \times \mathbf{N}$ to $\mathbf{N}$ given by $f(x, y) = x + y$ and $g(x, y) = x\,y$ are onto but not 1–1, where $\mathbf{N} = \{0, 1, 2, \cdots\}$.

5. Define Primitive recursive function. Show that the function $f(x, y) = x + y$ is primitive recursive function and hence find $f(2, 3)$.

6. Prove that a subset $S \neq \emptyset$ of group $(G, *)$ is a subgroup of G if and only if $a * b^{-1} \in S$, for all $a, b \in S$.

7. (a) Give an example of a semi group which is not a monoid.

   (b) Give an example of a monoid which is not a group.

8. Describe Konigsberg bridge problem. Does solution of the problem exists.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Show that P is equivalent to the following formulae,

   (a) $\neg\neg P$

   (b) $P \wedge P$

   (c) $P \vee P$

   (d) $P \vee (P \wedge Q)$

   (e) $P \wedge (P \vee Q)$

   (f) $(P \wedge Q) \vee (P \wedge \neg Q)$

   (g) $(P \vee Q) \wedge (P \vee \neg Q)$.

10. (a) Define quantifier and negation of quantifier statements.

    (b) Construct the truth table of $(Q \wedge (P \rightarrow Q) \rightarrow P)$.

11. (a) Show that $A \subset B \Leftrightarrow A \cap B = A$

(b) Show that $A \subset B \Leftrightarrow A \cup B = B$

(c) Show that $A \subset B \Leftrightarrow \sim B \subset \sim A$.

(d) If $S = \{1, 2, 3, 4, 5\}$, find the non-empty disjoints sets A_1 and A_2 such that $A_1 \cup A_2 = S$.

12. (a) Draw the Hasse diagram of the partially ordered set $(\{1, 2, 3, 6, 12\}, |)$, where '|' denotes divides.

(b) Under what condition on X, $(\rho(X), \subseteq)$ is totally ordered?

(c) Give an relation which both a partially ordering relation and an equivalence relation on a set.

13. Let $f : X \to Y$ and $g : Y \to X$. Prove that $g = f^{-1}$ if and only if $g \circ f = I_X$ and $f \circ g = I_Y$.

14. State and prove the fundamental theorem of group homomorphism.

15. (a) Prove that in any graph, the number of odd degree vertices is always even.

(b) Draw the digraph G corresponding to the adjacent matrix $\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$.

DE–762 **23**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2014.

OBJECT ORIENTED PROGRAMMING AND C++

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the questions.

1. What are the features of object oriented programming? Explain.

2. Explain the operators in C++ that cannot be overloaded.

3. List some of the special characteristics of constructor.

4. Write short note on inline function.

5. What is 'this' pointer? How it is available to member function of a class?

6. Parameter of a copy constructor can be passed by value? Discuss.

7. Explain operator overloading with example.

8. What is the visibility mode in inheritance? Explain.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the object oriented programming features in C++.

10. Explain the idea of classes, data abstraction and encapsulation.

11. Explain explicit constructors, parametrized constructors, and multiple constructors with suitable example.

12. Write a program using friend functions for overloading <<and>> operators.

13. Explain the different types of polymorphism.

14. Explain different types of inheritances with suitable example of each one of them.

15. Write a C++ program to illustrate the use of virtual functions.

DE–763 **24**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2014.

OPERATING SYSTEMS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Write short note on early operating systems.

2. Write short notes on Race conditions.

3. What are the various process scheduling concepts?

4. What is an Interrupt? Explain interrupt handlers.

5. Explain the scheme for detecting deadlock.

6. Describe paging with illustrative example.

7. Write short notes on buddy system.

8. Explain the concepts involved in maintaining the file system security.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the components of an operating system.

10. List out the various process states and briefly explain with a state diagram.

11. What do you mean by processor scheduling? Explain the various levels of scheduling.

12. Discuss any two methods of deadlock prevention.

13. Explain the various contiguous memory allocation strategies with suitable examples.

14. Discuss the concept of demand paging.

15. Explain any one directory structure.

DE–764 **25**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (Five Year Integrated) DEGREE EXAMINATION, DECEMBER 2014.

RELATIONAL DATABASE MANAGEMENT SYSTEM

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

All questions carry equal marks.

1. Write short notes on Data base scheme.

2. Explain the Data Manipulation Language (DML).

3. Mention the strengths of relational model.

4. Explain the data-types of oracle.

5. Write short notes on Sub Queries.

6. Give the purpose of secondary key.

7. How to select distinct values through SQL? Illustrate.

8. Explain the techniques of modifying tables.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss in detail about features and advantages of database management systems.

10. With sketch describe the DBMS architecture.

11. Explain the following in relational model.

    (a) Data definition facilities.

    (b) Data manipulation facilities.

12. What is E–R model? What are the components used to draw an E–R diagram? What are the advantages and disadvantages?

13. Describe about join theory in SQL.

14. Discuss about the languages supported by oracle precompiler in embedded SQL.

15. Create a sales order database of 2 tables Sales_ order_ and sales_ order_ detail with the following fields-

    Sales_ order

    Sales_ order_ detail

    S_ order_ no

    s_ order_ no

    S_ order_ date

    product_no

    Client_no

    Qty_oredered

    Salesman_no

    product_rate

    Dely_date

    Order_ status

    Set relationship between 2 tables.

DE–765 **31**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2014.

COMPUTER NETWORKS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Compare the OSI and TCP/IP reference models.

2. Explain how twisted pair cable is used in data communication.

3. Discuss on error correcting codes.

4. Explain about collision free protocols.

5. Explain congestion control in virtual circuit subnets and in datagram subnets.

6. Explain RARP.

7. Discuss crash recovery strategies.

8. Write a note on audio and video compression.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss in detail on wireless transmission.

10. Discuss on connection oriented networks.

11. How petrinet models and FSM models are used in protocol verification? Explain in detail.

12. Explain hierarchical and broad cast routing.

13. Discuss the role of tunneling and fragmentations in internet working.

14. Describe in detail about TCP.

15. Explain the following :

    (a) DNS and SNMP

    (b) RSA algorithm

DE–767 **33**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2014.

SOFTWARE ENGINEERING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. What is computer software? Discuss the importance of software.

2. Discuss the role of management in software development.

3. Write a short note on software project estimation.

4. What is risk identification? Discuss.

5. Discuss on quality standards.

6. Discuss on prototyping.

7. What is unit testing? Discuss.

8. Write a short note on transform mapping design.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. List the different software development process models. Discuss any two models.

10. Discuss on the following :

    (a) Software metrics

    (b) Software maintenance cost.

11. Write a short note on software configuration management.

12. Discuss on the following :

    (a) Planning for software assurance

    (b) Risk management

    (c) Software reviews.

13. Write short note on the following :

    (a) Structured analysis

    (b) Requirement specification analysis.

14. (a) What is software validation? Discuss.

    (b) Discuss on block box testing.

15. Write detailed notes on concepts, principles and module level concepts in software design.

DE–768 **34**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2014.

COMPUTER GRAPHICS AND MULTIMEDIA

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write an overview about I/O devices.
2. Illustrate with an example, explain DDA line drawing algorithm.
3. Discuss about Cohen Sutherland line clipping algorithm.
4. Describe about Bezier curves with examples.
5. Explain any two methods involved in hidden surface removal.
6. Give detailed description about texture mapping.
7. Write detailed notes on authoring tools.
8. Describe the concept of virtual reality with examples.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Give detailed description about Bresenham's and Midpoint circle drawing algorithm.

10. Explain window viewport mapping and 2D transformations.

11. Discuss on various color models.

12. Explain about video compression with examples.

13. Describe about video conferencing and applications of multimedia.

14. Write detailed notes on components of multimedia.

15. Explain and multimedia project development parallel and perspective projections with suitable diagram.

DE–769 **35**

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2014.

UNIX AND SHELL PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

All questions carry equal marks.

1. Explain the salient features of UNIX operating system.

2. Write syntax options and uses of following UNIX commands.

   (a) stty

   (b) cat

   (c) tar

   (d) gzip.

3. Discuss about networking commands of UNIX.

4. (a) Explain about filter and pipes. (5)

   (b) How will you perform comparison of files? (3)

5. Discuss on mathematical, string and user defined functions.

6. Write notes on sorting and translating characters in shell programming.

7. Discuss about argument types and argument validation in korn shell program.

8. Describe about file management commands in C shell program.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

All questions carry equal marks.

9. (a) Write notes on Vi editor. (5)

(b) Discuss about Unix utilities. (10)

10. (a) Explain the following commands. (10)

(i) tail

(ii) pg

(iii) grep

(iv) join

(v) comm.

(vi) diff

(vii) awk

(b) Write about job control in UNIX operating system. (5)

11. Write shell script to perform the following :

(a) Concatenation of files

(b) Cut and paste

(c) Comparing files

(d) Counting the no. of lines, words and characters.

12. (a) Write notes on sed and awk. (5)

(b) Discuss about eval command and command execution process in korn shell. (5)

(c) Explain positional parameters and special parameters in korn shell. (5)

13. (a) Explain C shell features. (8)

(b) Write short notes on debugging of korn shell scripts. (7)

14. (a) Discuss on start up and shut down scripts of C shell.

(b) Write about environmental variables of C shell program.

15. (a) Describe about commands that are used for file management in C shell. (10)

(b) Write notes on the following : (5)

(i) umask

(ii) chmod

(iii) fstat

(iv) rmdir

(v) finger.

DE–770 **41**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2014.

VISUAL PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write notes on handling menus and dialogs in visual basic.

2. How to create a control? Explain with example.

3. Discuss on file system controls with relevant examples.

4. How to access database using any one data control? Explain.

5. Explain VC++ components using suitable examples.

6. Describe the procedure of importing VBX controls.

7. Discuss on mouse and keyboard events with examples.

8. Explain the concept of object linking and embedding in visual basic.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the fundamentals of programming in visual basic, objects and instances in VB.

10. How to create graphics for application? Illustrate with examples.

11. Explain document view architecture and MFC file handling.

12. Discuss the concept of handling events in VC++.

13. Describe the procedure of reading and writing documents.

14. Write detailed notes on user interface classes and DLL in VC++.

15. Explain database management with ODBC using relevant examples.

DE–772 **43**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2014.

WEB TECHNOLOGY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the usage of list and frames in HTML with example.

2. What is the purpose of bound property? Illustrate with example.

3. Explain the advantages of Java Beans.

4. How do you read servlet parameters? Explain with example.

5. Discuss about Tom Cat web server.

6. Explain in detail about debugging a JSP application.

7. Write a note on anatomy of a JSP page.

8. Discuss about application specific database actions.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Briefly explain about document object model and XML processors.

10. Discuss in detail about Bean developer kit.

11. What is introspection? How is it used in Java Beans? Discuss in detail.

12. Explain the following :

    (a) JSDK

    (b) The javax. servlet HTTP package.

13. How do you deal the syntax errors in JSP? Explain about dealing with runtime errors.

14. List the problems with servlets. Explain the anatomy of a JSP page.

15. With an example, explain about accessing a data base from a JSP page.

DE–773 **44**

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2014.

DATA WAREHOUSING AND MINING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define datamining and discuss the classification of data mining systems.

2. Discuss the characteristics of data warehouse.

3. Explain the role of association rules in datamining.

4. Briefly discuss the dimensional modeling of data.

5. Explain the chi-squared statistics based regression analysis.

6. Compare cluster with classification technique.

7. Explain the K-means algorithm for clustering.

8. Discuss in detail about the web knowledge mining taxonomy.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Clearly explain about data pre processing in data mining.

10. (a) Briefly discuss datamining query language. (6)

(b) Explain the role of fuzzy sets and logic in datamining. (9)

11. Explain Bayes theorem for classification with an example.

12. (a) Explain the single link algorithm for clustering. (9)

    (b) Explain the BEA for clustering. (6)

13. (a) Explain the genetic algorithm based clustering technique. (6)

    (b) Explain any one of the categorical clustering algorithms. (9)

14. Explain in detail web content mining using crawlers and harvest systems.

15. Discuss any one web mining application.

DE–774 **45**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2014.

DISTRIBUTED COMPUTING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Briefly discuss on internet protocols.

2. What is networking and internetworking? Discuss.

3. Discuss on external data representation and marshalling.

4. Write a note on communication between distributed objects.

5. Write a brief note on processes and threads.

6. Give an overview of OS architecture.

7. What is the purpose of DNS? Discuss.

8. Briefly discuss on flat and nested distributed transactions.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss briefly on architectural and fundamental models.

10. Explain briefly, network principles.

11. What is Remote procedure call? Discuss.

12. Write a brief note on Java RMI.

13. Briefly discuss on X.500 directory service.

14. What is distributed mutual exclusion? Discuss.

15. Write a brief note on distributed deadlocks.

DE–8862 **16**

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2014.

LAB I — C PROGRAMMING

Time : Three hours Maximum : 100 marks

ONE question to be selected by lot system.

Each sub division carries 50 marks.

1. (a) Write a C program to count the occurrence of number in a given set of numbers (For example set of number 5, 5, 4, 4, 5 is 4 occurs 2 times, and 5 occurs 3 times)

   (b) Write a C program to check whether the given number is perfect number or not. (28 is a sum of its divisors (1 + 2 + 4 + 7 + 14 = 28).

---------- Cut here ----------

2. (a) Write a C program to sort the numbers in Ascending and Decending order using array :

   (For example 6, 7, 3, 5, 4 is

   Ascending order : 3, 4, 5, 6, 7 and

   Descending order : 7, 6, 5, 4, 3)

   (b) Write a C program to check the given string is palindrome or not.

   (For example MADAM is palindrome).

3. (a) Write a C program for matrix multiplication using function.

   (b) Write a C program to count the number of vowels in the given string.

---------- Cut here ----------

4. (a) Write a C program to read 10 values into an array variable. Use pointers to locate and display each value.

   (b) Write a C program to convert a given number into words for numbers 1 to 5.

   (For example 1 to one, 2 to two and 5 to five).

---------- Cut here ----------

5. (a) Write a C program to solve a Quadratic equation $ax^2 + bx + c = 0$.

   (b) Write a C program to check the given word is palindrome or not.

   For example MADAM IS palindrome)

---------- Cut here ----------

6. (a) Write a C program to implement all string operations using SWITCH statement.

   (b) Write a C program to find the factorial of a given number using recursion.

---------- Cut here ----------

7. (a) Write a C program to prepare student marklist using UNION and Structures.

   (b) Write a C program to reverse the string without using string function.

8. (a) Write a C program to arrange the given 10 names in alphabetical order.

(b) Write a C program for finding the second maximum of 5 given numbers.

DE-8863 **17**

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2014.

DATA STRUCTURES LAB

Time : Three hours Maximum : 100 marks

ONE question should be given to each, candidate by lot system.

Each sub-division carries 50 marks.

1. (a) Write a C program to copy contents of one file to another file.

   (b) Write a C program to implement insert and delete operation on linked list structure.

---------- Cut here ----------

2. (a) Write a C program to sort no's in ascending order and display values before and after sorting.

   (b) Write a C program for linked list implementation of queue operations.

---------- Cut here ----------

3. (a) Write a C program for linked list implementation of queue operations.

   (b) Write a menu driven program in 'C' to find an element using linear and binary search methods.

4. (a) Write a 'C' program to implement insert and delete operations on linked list structure.

(b) Write a 'C' program to sort a elements using insertion sort.

---------- Cut here ----------

5. (a) Write a 'C' program to sort elements using selection sort.

(b) Write a 'C' program to copy contents of one file to another file.

---------- Cut here ----------

6. (a) Write a 'C' program to implement push and pop operation on stack.

(b) Write a 'C' program for linked list implementation of queue operations.

---------- Cut here ----------

7. (a) Write a 'C' program to sort nos in descending order and display values before and after sorting.

(b) Write a 'C' program to evaluate the given mathematical expression using stack.

---------- Cut here ----------

8. (a) Write a 'C' program for linked list implementation of queue operation.

(b) Write a 'C' program to sort a set of elements using insertion sort.

DE–8864 **26**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2014.

C++ LAB

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by lot system. Each sub-division carries 50 marks.

1. (a) Write a program to process student marks with the help of classes. The class has private variables, name, mark 1, mark 2, mark 3. It has two member functions – getdata ( ) to get input, result ( ) to print the results. All subjects mark must be > = 50 for pass otherwise fail.

   (b) Create a class 'COMPLEX' to hold a complex number. Write a friend function to add two complex numbers. Write a main function to add two COMPLEX objects.

---------- Cut here ----------

2. (a) Using dynamic constructors, write a C++ program to concatenate two given strings.

   (b) Create a 'DISTANCE' class with : feet and inches as data members, member function to input distance, member function to output distance, member function to add two distance objects. Write a main function to create objects of DISTANCE class. Input two distances and output the sum.

3. (a) Create a class DONOR that contains donor number, donor name, age, address, sex, blood group. Write a program using C++ to print out the number, name and address of the donors for the following categories :

   (i) Blood donors having O+ blood group

   (ii) Blood donors in the age group between 20 to 30

   (iii) Female donors with A+ blood group in the age between 25 to 35.

   (b) Write a C++ program to sort an array of integers in ascending order using function called exchange ( ) which accepts two integer arguments by reference.

---------- Cut here ----------

4. (a) Write a program using polymorphism to calculate the square of any two numbers of type integer, float, double and long.

   (b) Create a class called 'EMPLOYEE' that has EMPCODE and EMPNAME as data members, member function getdata ( ) to input data, member function display ( ) to output data. Write a main function to create EMP, an array of EMPLOYEE objects. Accept and display the details of atleast 6 employees.

5. (a) Raising a number n to power p is the same as multiplying by itself t times. Write a function called power ( ) that takes a double value for n and an int value for p and returns the result as double value. Use default argument of 2 for p, so that if this argument is omitted, the number will be squared. Write a main ( ) function that gets values from the user to test this function.

(b) Write a C++ program that uses a single file for both reading and writing the data.

---------- Cut here ----------

6. (a) Create a class called TIME that has integer data elements for hours, minutes, seconds. The constructors should initialize these data elements to specified value, if given and otherwise to 0. A member function should display it, in 11:50:45 format. The final member function should add two objects of type time passed as arguments.

(b) Create a 'MATRIX' class of size m × n. Overload the '+' operator to add two MATRIX objects. Write a main function to implement it.

---------- Cut here ----------

7. (a) Assume that you want to generate a table of multiples of any given number. Write a program that allows the user to enter the number and then generates the table, formatting it into ten columns and 20 lines.

(b) Write a C++ program to illustrate multilevel inheritance.

8. (a) Write a menu driven C++ program to add, substract given two matrices of order m × n defined in class using operator overloading.

(b) Write a C++ program to sort an array of integers in descending order using a function called exchange ( ) which accepts two integer arguments by reference.

DE-8865 **27**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated)
DEGREE EXAMINATION, DECEMBER 2014.

RDBMS — LAB
(RELATIONAL DATABASE MANAGEMENT SYSTEMS)

Time : Three hours Maximum : 100 marks

ONE question should be give to each candidate by lot system.

1. Create an employee database with the following fields :

   Employee name, Data of birth, Date of Joining, Department, Designation, Grade (A/B/C), Basic pay, Address Line1, Address Line 2, Place.

   Write programs in ORACLE to perform the following :

   (a) List of employees who are coming from Madurai and earning a Basic pay more than Rs.15,000/-

   (b) Increase the basic pay by 20% for Grade 'A' employees.

   (c) Increase the basic pay by Rs.5,000/- for employees whose designation is 'Manager'.

2. Create a patient data file with the fields.

Patient name, Patient-Type (IN/OUT), No.of days admitted (for in-patients) Amt per day (in-patient), Medicine Amt, Blood test taken (Yes / No), Blood test charge, Amount paid.

Write an Oracle program to perform the following :

(a) For out-patients the medicine-amount and blood test charge (if any) must be added and stored in amount paid field.

(b) For in – patient

Amount paid = (No. of days admitted × Amt per day) + Medicine amount + Blood Test Charge (if any).

(c) Generate the list in the following format :

Patient name	Type	Amount paid
_	_	_
_	_	_
_	_	_

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3. Create a database with the following fields :

Account No, Customer name, Opening balance, Closing balance, Tran type, Tran amount, Write an Oracle program to perform the following :

(a) If Tran type = "Deposit"

Closing balance = Opening balance + Tran Amount.

(b) If Tran type = "Withdrawal"

Closing balance = Opening balance – Tran Amount

if closing balance is a negative value, then assign opening balance to closing balance.

4. Create an employee database with two tables employee (master table) and overtime (transaction table).

Employee	Overtime
Emp. No.	Emp. No.
Emp. Name	OT date
Department	OT hours
Designation	
Basic pay	
Cross pay	
Net Pay	

for each employee, enter atleast 2 to 3 records in overtime table.

Compute the allowances and deductions as follows :

DA = 43 % of basic pay

HRA = 10% of basic pay

OT amount = sum of OT hours per employee

* 500 (per OT day)

Gross salary = Basic pay + DA + HRA + OT Amount

PF = 10% of basic pay

IT = 12% of basic pay

Net pay = Gross pay – PF – IT

Generate the list in the following format :

Employee name	Gross pay	Net pay
–	–	–
–	–	–
–	–	–

5. In a hostel, a data file is maintained with the following fields :

Student name, Amount to be paid, Days absent, Days delayed, Amount paid.

Write a program in oracle to perform the following :

Amt to be paid = Rs.2,500/- for all students if days absent × 5, then reduction is computed as reduction = (days absent – 5) ∗ 80. The penalty for each day they delay is Rs.20/- per day.

Penalty = Days Delayed ∗ 20.

Amt paid = 2500 – Reduction + Penalty

Generate the list as follows :

Student Name	Days Absent	Days delayed	Amount paid
–	–	–	–
–	–	–	–
–	–	–	–

6. Create a loan datafile with the following fields :

Loan No.	Customer Name	Principle amt
No of years	Rate of interest	EMI (equated monthly installment)

Calculate Interest = $\frac{P * N * R}{100} * 12$

EMI = $\frac{(\text{Principle (P)} + \text{Interest})}{\text{N} * 12}$

Display the details as follows :

Loan No.	Customer Name	EMI
–	–	–
–	–	–
–	–	–

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7. Create two files invoice head (master) and invoice details (transaction) with the following fields :

Invoice head	Invoice detail
Invoice No.	Invoice No.
Customer name	Prod ID
Invoice date	Qty sold
Invoice amount	Prod price

Relate both the files using 'Invoice No'. For each invoice number in master, enter 3 or more records in invoice details. For each invoice number. Compute the sum of (Qty sold * prod price) for all products that come under one invoice number and replace the respective invoice amt field with the computed sum.

Finally generate the list as follows :

Invoice No.	Date	Invoice Date
–	–	–
–	–	–
–	–	–

8. Create a loan datafile with the following details :

Loan No, Customer name, EMI, Due date, Paid date, Amt paid.

Calculate the delayed days as paid date – due date

Calculate the penalty as No. of delayed days * Rs.100/-

Amount paid = EMI + Penalty

Display the details as

Loan No.	Customer Name	EMI	Amt paid
–	–	–	–
–	–	–	–
–	–	–	–

---------- Cut here ----------

9. Create a library database with the following fields :

Member No, Member name, Book ID, Lending date, Due date, Returning date, Fine amount

Member No, Member name, Book ID, Lending date, Due date, Returned date, Fine Amount,

Write an oracle program to perform the following :

Due date = Lending date + 7.

If returned date and due date are same.

No penalty is calculated.

If returned date > Due date, an amount of. 50 (paise) is collected as penalty.

* Generate the list as follows :

Member name	Fine amount
–	–
–	–
–	–

10. Create a students database with the following details : Register Number, Student name, Semester No, Mark 1, Mark 2, Mark 3.

Write an oracle program to print the data in the following format :

Sl.No.	Name	Mark 1	Mark 2	Mark 3	Total	Average	Class
–	–	–	–	–	–	–	–
–	–	–	–	–	–	–	–
–	–	–	–	–	–	–	–

Total = Mark 1 + Mark 2 + Mark 3

$$\text{Average} = \frac{\text{Total}}{3}$$

Result = "pass", if passed in all subjects Mark 1, Mark 2 and Mark 3.

For pass , minimum marks is 50.

Class = I if average > = 60

II if 50 < average = 60

Fail if average < = 50.

DE–8866 **36**

DISTANCE EDUCATION

M.Sc. (SE) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2014.

INTERNET AND JAVA – LAB

Time : Three hours Maximum : 100 marks

ONE question to be selected by lot system.

1. (a) Write a VB script to do the following :

   (i) Check the given password

   (ii) Change the existing password.

   (b) Write a Java script to create a window by using the confirm message.

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2. (a) Write a VB script to create a Calender for given month and year.

   (b) Write a swing program to create the tabbed panels.

---------- Cut here ----------

3. (a) Write a Java program using applet to display the different colors and fonts.

   (b) Write a Java script to create a color palette and display the text in the color chosen from the palette with proper background color.

4. (a) Write a Java script to create a order form to select the house articles.

(b) Write a Java program to create a color as the back ground.

---------- Cut here ----------

5. (a) Write a Java program using applet to display any three images when three buttons in the border layout are clicked. The image should be displayed at the center.

(b) Write a swing program to create buttons with

(i) Tool tip text

(ii) Image

(iii) Border

(iv) Short cut key.

---------- Cut here ----------

6. (a) Write a Java script to create a order form to select the house articles.

(b) Develop a single page advertisement for a shop to be opened shortly.

7. (a) Write a Java program using applet to create the frames and its controle.

(b) Using atleast 20 HTML tags, create a screen with a string "WEB DESIGN".

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8. (a) Design and publish a web page for a college.

(b) Write a Java program using applet to display the dialogue menu in applet.

DE–8867 **37**

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2014.

UNIX AND SHELL PROGRAMMING LAB

Time : Three hours Maximum : 100 marks

ONE question to be selected by lot system.

Each subdivision carries 50 marks.

1. (a) Write a Menu Driven Shell program for the following :

   (i) List of files

   (ii) Processes of users

   (iii) Today's date

   (iv) Users of system

   (v) Quit to UNIX.

   (b) Write a shell program using 3 arguments to take the pattern as well as input and output file names. If the pattern is found, display "Pattern found" else display "Error message". Also check if correct number of arguments are entered.

2. (a) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it :

   (i) File existence

   (ii) File readable

   (iii) File writable

   (iv) Both readable and writable.

   (b) Write a shell program to sum up the series

   1/1! + 1/2 ! + 1/3! + ———————— + 1/10!

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3. (a) Write a Menu Driven shell program to perform the following tasks

   (i) Enter the sentence in a file

   (ii) Search a given whole word in an existing file

   (iii) Quit.

   (b) Write a shell program to prepare electric bill for domestic consumers.

   For first 100 units – Rs. 0.75/unit

   For next 100 units – Rs. 1.50/unit

   Above 200 units – Rs. 3.00/unit

   Prepare the Electricity bill in the following format :

   Customer no ————————

   Customer name ————————

   Previous reading ————————

   Current reading ————————

   Units consumed ————————

   Charge ————————

4. (a) Write a program for four-function calculator by using yacc command.

(b) Write a shell program which periodically monitors the disk for the existence of a file and then executes the program once the file has been located. (using UNTIL statement).

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5. (a) Write a shell program which accepts the name of a file from the standard input and output then perform the following tests on it :

(i) Enter the 5 names in a file

(ii) Sort the names in existing file

(iii) List unsorted and sorted file

(iv) Quit.

(b) A Hospital maintains patient details. Write a shell program to list the following.

(i) Patients in a blood groupwise

(ii) Patients in age between 20 to 30

(iii) Exit.

---------- Cut here ----------

6. (a) Write and execute a shell program to do the following :

(i) Convert given string in center of given row in BOLD, normal or reverse video.

(ii) Write given string at given row, column in BOLD, Normal and reverse video.

(iii) Search a given string using awk with various options.

(b) Write and execute a shell program to send mail to groups of users by extracting their Ids from/etc/group file.

7. (a) Write and execute a shell program to generate leave status report of an organization.

(b) Write and execute a shell program to do the following :

(i) Convert given string in center of given row in BOLD, normal or reverse video.

(ii) Write given string at given row, column in BOLD, Normal and reverse video.

(iii) Search a given string using awk with various options.

---------- Cut here ----------

8. (a) Write a shell program which accepts the name of a file from the standard input and output then perform the following tests on it :

(i) Enter the 5 names in a file

(ii) Sort the names in existing file

(iii) List unsorted and sorted file

(iv) Quit

(b) Write and execute a shell program to send mail to groups of users by extracting their Ids from/etc/group file.

DE–8868 **46**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, DECEMBER 2014.

VISUAL BASIC AND VC++ LAB

Time : Three hours Maximum : 100 marks

Marks Breaks –Up

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

Examiner should select and give ONE question to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Using a control array, create a simple calculator.

   (b) Write a Visual C++ program to generate a status bar and show the status of caps lock, num lock and scroll lock in it.

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2. (a) Develop a data report using employee table with the following:

   (i) Display all employee details

   (ii) Display all employee details in each department and display total salary in each department.

   (iii) Display all employee details which starts with the employee name specified in text box.

   (b) Write a VC++ program to handle windows messages in MFC program.

3. (a) Write a VB program to perform the following operation in a record of random access file

   (i) insert

   (ii) delete

   (iii) Search.

   (b) Write a VC++ program to find out whether a mouse is attached or not; and if attached how many buttons are present.

---------- Cut here ----------

4. (a) Create a form to design the examination application form of a college using Text box, combo box, command button. Assume your own model for the exam form.

   (b) Write Visual C++ program to fill the background of the client area with a bitmap.

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5. (a) Develop a data report using student personal table to display the following:

   (i) Display particulars of all students

   (ii) Display total SC male and SC female students

   (iii) Display percentage pass of SC students.

   (b) Write a Visual C++ program to create a list box in window

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6. (a) Using built-in AX control, develop the windows Notepad with file and edit menu operations and display the floating menu whenever necessary.

   (b) Write a VC++ program to manage window sizes using MFC.

7. (a) Using Active X control create a Textbox that accepts only numeric value with the following properties:

   (i) background of the textbox

   (ii) resize the textbox

   (iii) text property of the textbox

   (iv) resize the textbox at the standard application without using properties.

   (b) Write a VC++ program, to find whether mouse is attached or not. If attached, how many buttons are attached.

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8. (a) Create a VB form with Text box, Combobox and Command button and do the following operation.

   (i) Add the university name in the Combobox at the run time

   (ii) Search and delete the particular university in the Combobox

   (iii) Sort the university names in the alphabetical order.

   (b) Write a VC++ program to get the status of the shift and toggle keys using MFC.

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9. (a) Create a VB application with a DriveListBox, DirListBox

   (i) Select exe file and execute it.

   (ii) Select the picture and load it to the form.

   (iii) Filter the file in the file list box according to the extension chosen in the combo box.

   (b) Write a VC++ program to fill background of the client area with a bitmap.

10. (a) Create a table hospital with the following fields:

Patient number, name, address, blood group, disease.

Using Ax data access object develop a hospital management system with the following operations

(i) Insert and delete a record

(ii) Search and modify the record

(iii) Display all the records whose blood group is O +ve

(b) Write a VC++ program to create a window of desired size using MFC.

DE–8869 **47**

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, DECEMBER 2014.

WEB TECHNOLOGY LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot systems.

Break-up of marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

1. (a) Design a webpage using HTML code to display the details of Bus Reservation Form.

   (b) Write a Servlet program to display "Hello World".

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2. (a) Design a web page using HTML code to display the front page of library information.

   (b) Write Servlet program to send Email message.

3. (a) Design a web page using HTML code that moves your name randomly around on a page.

(b) Write a XML document to display your bio-data. Write an XSL style sheet and attach that to the XML document. Validate the document using DTD or XSD.

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4. (a) Write a JSP program to display Super Market billing details using JDBC.

(b) Write an ASP program to prepare Employee pay bill using Java Script.

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5. (a) Write Servlet program to create a chatting application.

(b) Write an ASP program to prepare student performance evaluation document using Java Script.

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6. (a) Develop an application for JSP–servlet communication.

(b) Design a web page using HTML code to display the details of a college.

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7. (a) Write a JSP program to display student details using JDBC.

(b) Design a webpage using HTML code to display the details of the places you visit.

8. (a) Create an Image animation using Java Script.

(b) Create a HTML table with several rows and columns and split them using Rowspan and Colspan.